DRUGS · NASAL PREPARATIONS

Topical nasal decongestants

Xylometazoline and oxymetazoline are alpha-adrenergic agonists that shrink the nasal mucosa within minutes. They relieve congestion in acute colds and sinusitis, open the nose before examination and surgery, help in epistaxis, and are a diagnostic tool when a fixed obstruction is suspected. Used for more than a few days, they cause rebound congestion and rhinitis medicamentosa.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

TOPICAL NASAL DECONGESTANTS

	SUMMARY
Agents	Xylometazoline 0.1% (adults) and 0.05% (children); oxymetazoline 0.05%; phenylephrine (in combinations)
How they work	Alpha-1 and alpha-2 adrenoceptor agonists: venous sinusoid constriction in the turbinates
ENT uses	Short-term relief of congestion in colds and acute sinusitis; preparation for nasendoscopy and procedures (often with lidocaine); epistaxis (on pledgets); before nasal surgery; Eustachian tube symptoms in air travel; testing whether obstruction is mucosal (shrinks) or structural (does not)
Adult dose	Xylometazoline 0.1%, 1 to 2 sprays into each nostril up to three times daily, for no more than 7 days
Child dose	In the UK, not recommended under 6 years; 6 to 11 years xylometazoline 0.05%, 1 spray up to twice daily for up to 5 days
Avoid	Within 2 weeks of an MAOI; after trans-sphenoidal surgery; dry rhinitis; narrow-angle glaucoma; caution in hypertension, heart disease, hyperthyroidism

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Mechanism and rebound

They constrict the capacitance vessels of the turbinates. With repeated use, receptors downregulate and the mucosa congests again as the effect wears off (**rebound**), prompting more use. After about 5 to 10 days this becomes **rhinitis medicamentosa**: persistent congestion, a red, swollen, granular mucosa, and dependence on the spray.

3 CHAPTER THREE Using them in ENT

Indications and doses.

3.1 In the clinic and theatre

Applied with lidocaine as a spray or on cotton pledgets, they improve the view and comfort of nasendoscopy and procedures. In theatre, oxymetazoline or adrenaline pledgets reduce bleeding in endoscopic sinus surgery. In **epistaxis**, a pledget with decongestant before cautery is standard first aid.

3.2 Treating rhinitis medicamentosa

Stop the spray (one nostril at a time if needed), cover the withdrawal with a **nasal corticosteroid** for several weeks, and sometimes a short course of oral prednisolone; treat the original cause of congestion.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Rebound congestion and rhinitis medicamentosa.
- Local stinging and dryness.
- Systemic absorption: hypertension, palpitations and, in infants and young children, CNS depression and bradycardia (hence the age limits).

4.2 Interactions

MAOIs (hypertensive crisis), tricyclics, beta blockers (reflex hypertension with phenylephrine).

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism of **rebound** and rhinitis medicamentosa, and how to treat it.
- Decongestion as a test of mucosal versus structural obstruction.
- Age limits in children.

5.2 Common pitfalls

COMMON PITFALLS

- Allowing patients to use decongestant sprays for weeks.

- Using them in infants.

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